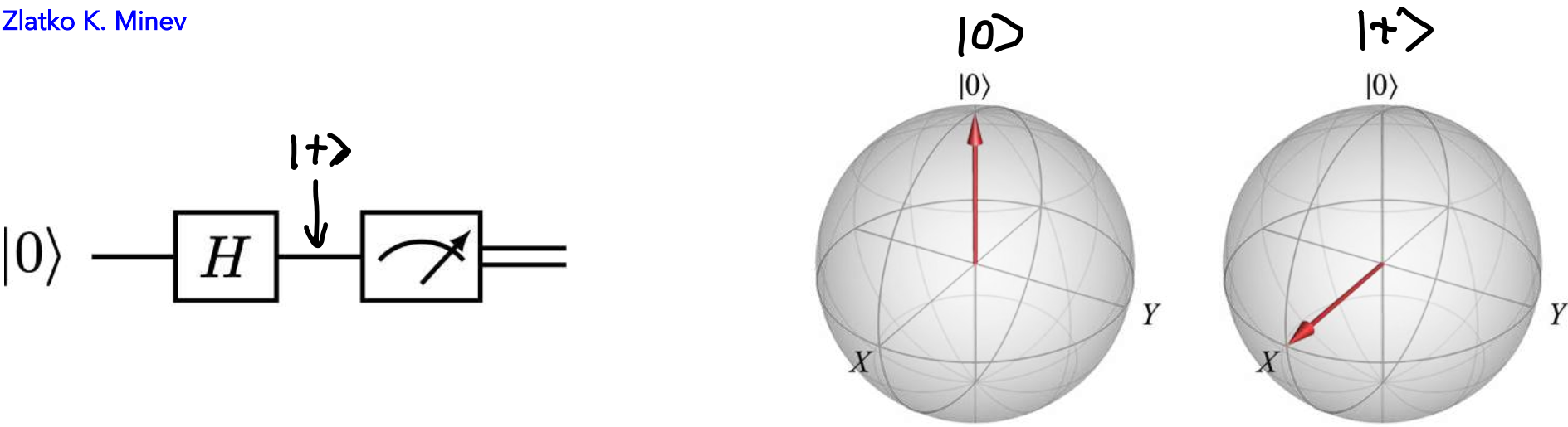


Introduction to quantum noise

Measurement theory & projection noise

Qiskit Global Summer School on Quantum Machine Learning
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Measurement: theory 101

The standard (von Neumann) measurement of a quantum system.
von Neumann measurement is efficient, strong, and projective

Measurement operator (observable) $\hat{M}$

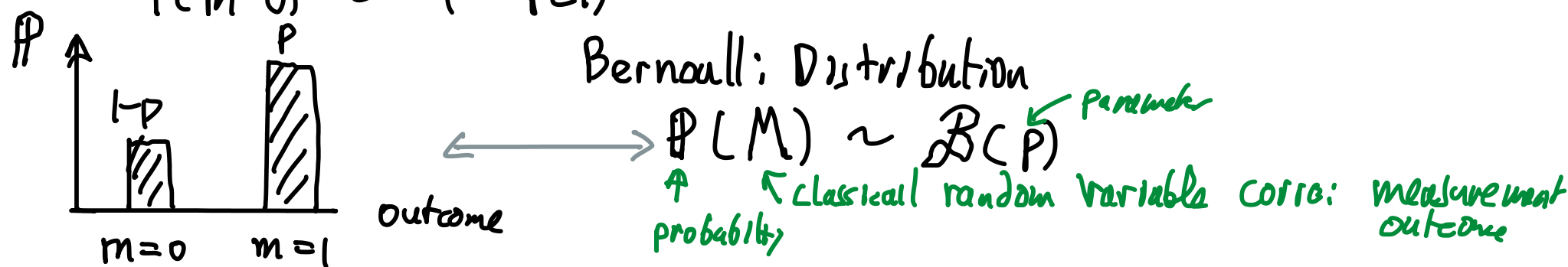
$|\psi\rangle$ $\xrightarrow{p}$ $\begin{matrix} 0/1 \\ \text{classical} \\ \hat{M} \end{matrix}$ $\xrightarrow{m \leftarrow \text{outcome}}$ $m=0 \rightarrow |0\rangle$
 $m=1 \rightarrow |1\rangle$

Probability of outcome
 $p(m=0) = \langle 0| \psi \rangle \langle 0| \psi \rangle$
 $p(m=1) = \langle 1| \psi \rangle \langle 1| \psi \rangle$

$\hat{\Pi}_0 = |0\rangle\langle 0|$
 $\hat{\Pi}_1 = |1\rangle\langle 1|$
 $\hat{\Pi}_0 + \hat{\Pi}_1 = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \hat{I}$
 $p(0) + p(1) = 1$

Is the qubit in $|1\rangle$?
 $\hat{M} = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$
 $= 0|0\rangle\langle 0| + 1|1\rangle\langle 1|$
 $= \sum_m m |m\rangle\langle m|$
 $m \in \{0, 1\}$ $\leftarrow$ possible outcomes
 $|m\rangle \in \{|0\rangle, |1\rangle\}$
 $\hat{M} = \sum_m m \hat{\Pi}_m$
 $\hat{\Pi}_m = |m\rangle\langle m|$
 $\hat{f} = \sum_m \hat{\Pi}_m$ (resolution)

$p(m=0) = \langle \psi | 0 \rangle \langle 0 | \psi \rangle = |\langle 0 | \psi \rangle|^2$
 $p(m=1) = \langle \psi | 1 \rangle \langle 1 | \psi \rangle = |\langle 1 | \psi \rangle|^2 = p$
 $p(0) + p(1) = |\langle 0 | \psi \rangle|^2 + |\langle 1 | \psi \rangle|^2 = |\psi|^2 = \langle \psi | \psi \rangle = 1$
 $p(m=0) = 1 - p(1)$



Statistics

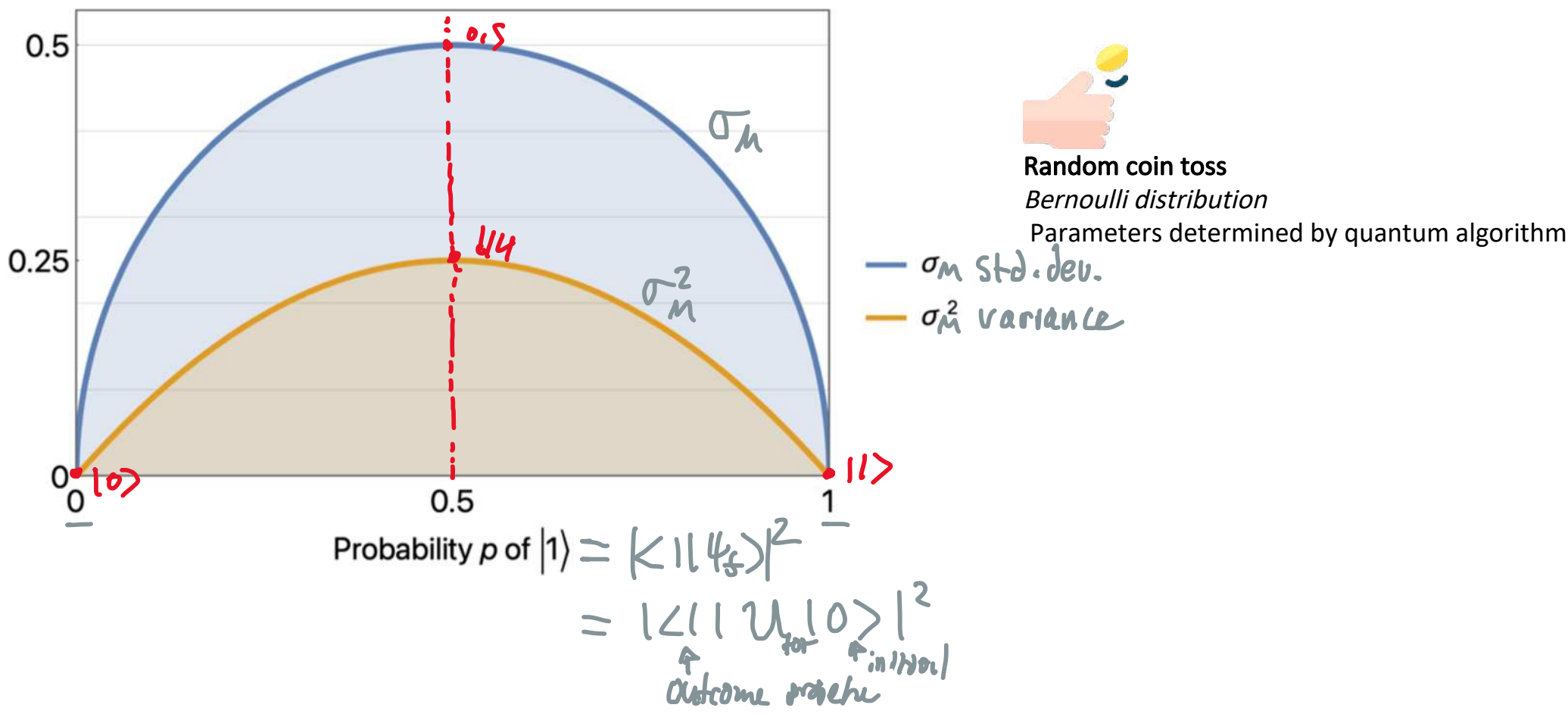
$\mathbb{E}[M] = \sum_m m P(m) = 0 P(m=0) + 1 P(m=1) = P$

$= \sum_m m \langle \hat{\Pi}_m \rangle$
 $= \sum_m m \langle |m\rangle\langle m| \rangle$
 $= \langle \sum_m m |m\rangle\langle m| \rangle$
 $= \langle \hat{M} \rangle$
 $\leftarrow$ Quantum

$V[M] = \mathbb{E}[M^2] - \mathbb{E}[M]^2 = \langle \hat{M}^2 \rangle - \langle \hat{M} \rangle^2$

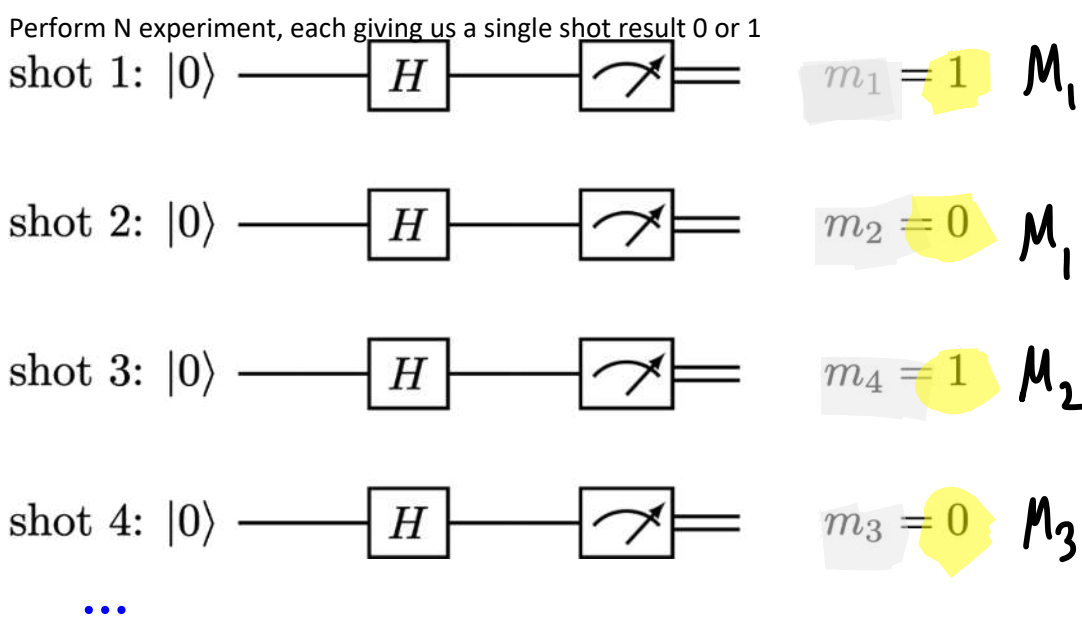
$\mathbb{E}[M^2] = \sum_m m^2 P(m)$
 $= \sum_m m^2 \langle |m\rangle\langle m| \rangle$
 $= \langle \sum_m m^2 |m\rangle\langle m| \rangle$
 $= \langle \hat{M}^2 \rangle$
 $= 0^2 \cdot (1-p) + 1^2 \cdot p$
 $= p$

$V[M] = p - p^2$
 $= p(1-p)$
 $= \sigma_M^2$
 $= 0$ if $p=0$
 $= 0$ if $p=1$



Projection noise and sampling error

Let's turn to the example of finite number of shots we execute for our experiment.



For 3 samples, there are 2^3 possible outcome sequences.

Sample mean
 $S := \frac{1}{N} \sum_{n=1}^N M_n$
 $= \frac{1}{3} (0 + 0 + 1) = \frac{1}{3}$
 $= \frac{1}{3} (1 + 0 + 1) = \frac{2}{3}$

$\mathbb{E}[S] = \mathbb{E}[\frac{1}{N} \sum_{n=1}^N M_n]$
 $= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[M_n]$
 $= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[M]$
 $= \mathbb{E}[M]$

$\mathbb{E}[aM] = a \mathbb{E}[M]$
 $\mathbb{E}[aM_1 + bM_2] = a \mathbb{E}[M_1] + b \mathbb{E}[M_2]$

$\hat{p} = \frac{1}{N} \sum_{i=1}^N M_i$ for agrob

$$\begin{aligned} \mathbb{V}[\hat{S}] &= \mathbb{V}\left[\frac{1}{N} \sum_{i=1}^N M_i\right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{V}[M_i] \\ &= \frac{p(1-p)}{N} = \frac{\mathbb{V}[M]}{N} \end{aligned}$$

Recall:

$$\begin{aligned} \mathbb{V}[aX] &= a^2 \mathbb{V}[X] \\ \mathbb{V}[aX + bY] &= a^2 \mathbb{V}[X] + b^2 \mathbb{V}[Y] + 2ab \text{Cov}[X, Y] \\ \text{Cov}[X, Y] &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \end{aligned}$$

error bar on expectation value

$$\sigma_{\hat{S}} = \sqrt{\frac{p(1-p)}{N}}$$

$\frac{p(1-p)}{N}$ ← Quantum projection noise
 $\frac{1}{N}$ ← finite sampling noise

